UX CASE STUDY

AI-Powered Intelligent Check-In System

Redesigning the arrival experience for Clients/ visitors, organizers, and hospitality teams through AI, empathy, and research-led design.

5 roles	User groups interviewed — Clients/visitors, organizers, receptionists, admins, hospitality team
34 sessions	User interviews, Surveys, Usability tests conducted over 8 weeks
~4 min	Average check-in time reduced from 11+ minutes to under 4 minutes in prototype testing
62%	Drop in front-desk interruptions in simulated workflow testing
91%	Task-completion rate achieved in moderated prototype testing across all personas
78%	Faster check-in time
80%	Reduction in receptionist workload
1	Connected system from entry to exit

01 · Project Overview

Large-scale events, client visit/meetings, corporate campuses and hospitality venues share a persistent friction point: check-in. What should be a seamless welcome moment is routinely spoiled by long queues, manual data entry, missed guest notifications, unorganised and overwhelmed front-desk staff — a problem felt equally by every stakeholder in the ecosystem.

I led end-to-end UX design for an AI-powered intelligent self-check-in platform, taking the product from a vague hypothesis — 'check-in should be smarter' — to a validated, high-fidelity prototype grounded in real-world research with five distinct user groups.

My Role	Duration	Methods	Platform
Principal Product Designer — Research, Strategy, IA, Interaction & Visual Design	16 weeks (Discovery → Prototype)	Contextual Inquiry · User Journey Mapping · Co-design · Usability Testing	Web App · Kiosk · Mobile · Admin Dashboard

02 · Problem Statement

Check-in looks like a two-minute administrative task. Beneath the surface, it is a multi-actor, multi-system challenge that reveals deep misalignments between how venues operate and what people actually need.

Initial Brief

The product brief I received was intentionally sparse: 'Explore how AI could modernize the check-in experience for events and workplaces.' There was no existing product, no defined scope, and no assumed solution — exactly the kind of problem space I find most rewarding to research.

Hypothesis

> **Working Hypothesis - We** believe that a significant portion of check-in frustration is caused not by the process itself, but by information gaps — guests not knowing what to expect, staff not knowing who is arriving, and organizers not having visibility into real-time attendance. AI-driven personalization and proactive communication could close these gaps.

The Problem Definition

Before conducting any interviews, I mapped a preliminary problem landscape to identify the ecosystem's stakeholders, their likely pain points, and the data flows involved. This helped me structure my research plan without front-loading assumptions.

Stakeholder ecosystem identified at outset:

- Visitors / Clients — people arriving at an event or location
- Event Organizers — responsible for attendee experience and logistics
- Receptionists / Front-Desk Staff — primary gatekeepers in the physical space
- Admins / Operations — manage system setup, data, and compliance
- Hospitality Team — ensure on-site experience quality (catering, rooms, AV, etc.)

> **Design Challenge How** might we design an AI-powered check-in experience that reduces wait times and staff burden, while making every guest feel recognized, informed, and welcomed — across diverse venue types and event scales?

03 · Research Planning

A problem this multi-stakeholder demanded a layered research approach. I designed a plan spanning eight weeks, combining qualitative depth with enough breadth to identify cross-role patterns.

Research Goals

- Understand the current end-to-end check-in journey from every stakeholder's perspective
- Identify the most painful moments and their root causes
- Surface unmet needs and latent opportunities — things people had stopped expecting to be fixed
- Understand how AI / automation was perceived by staff and guests (trust, anxiety, preference)
- Establish behavioural baselines: time-on-task, workarounds, error rates

Research Methods

Method	Participants	Why
Contextual Interviews	8 visitors and 38 clients feedback submission, 6 organizers, 2 receptionists, 4 admins, 4 hospitality staff	Rich narrative data; surface emotion, workarounds, and hidden complexity
On-site Shadowing & Observation	3 corporate campuses, 2 conference venues	See the real behaviour vs reported behaviour; capture environment constraints
Diary Study (Guests)	6 participants over 3 event cycles	Longitudinal view of pre-arrival anxiety, in-event experience, and post-event recall
Card Sorting (Staff)	8 admin & reception staff	Understand mental models around data fields and notification priority

Recruitment Strategy

I partnered to recruit participants across all five stakeholder groups. Across venues I conducted approximately 34 sessions (interviews, diary check-ins, and card sorts) over eight weeks.

04 · User Research

4.1 Client/Visitors & Guests

I conducted contextual interviews with visitors, verified the clients feedback, ranging from first-time visitors to frequent corporate campus visitors. I asked them to walk me through their most recent check-in experience in detail, narrating what they did, what they felt, and what they wished had been different.

Key Findings — Visitors

- Pre-arrival anxiety was near-universal: 6 of 8 participants admitted to arriving early specifically to buffer against expected check-in delays
- 'Invisible queue' frustration: visitors couldn't tell whether they were in the right queue or how long it would take — this uncertainty felt worse than the actual wait
- Name-spelling friction: 5 participants reported their names being mistyped or mispronounced, which they described as 'instantly deflating' — the opposite of a welcome
- Information asymmetry: guests often didn't know where to go after checking in — which floor, which room, which contact person if organiser is not available during checkin
- Privacy ambivalence: guests were comfortable sharing data for a smoother experience but wanted to understand why each data point was needed

> **Standout Quote — Visitor** "I arrived 20 minutes early specifically because I knew check-in would be chaos. And it still was. I just stood there staring at a paper list while the receptionist squinted at my surname."

4.2 Organizers

Six organizers were interviewed — three from corporate L&D teams running internal conferences, others are PMO department

Key Findings — Organizers

- Real-time visibility was the number-one unmet need: organizers were flying blind during events, relying on manual headcounts or hoping reception would tell them if something went wrong
- Last-minute attendee changes (additions, cancellations, substitutions) caused disproportionate stress — current systems had no graceful mechanism for on-the-day updates
- No-show prediction would be 'transformative': organizers could adjust catering, seating, and AV Guest wifi, food and beverages in real time if they had even rough no-show forecasting
- Branding consistency was important but hard to achieve across check-in touchpoints

> **Standout Quote — Organizer** "I have a dashboard for everything except the event/meeting itself. I have no idea in real time how many people have actually walked through the door. It's embarrassing."

4.3 Receptionists

Six receptionists were interviewed across three venues, covering both dedicated event receptionists and general front-desk staff who handled check-in as one of many concurrent tasks.

Key Findings — Receptionists

- Cognitive overload during peak arrival windows: the 15-minute window before an event started was described as 'controlled chaos' — searching databases, answering questions, handling walk-ins, and managing queues simultaneously
- The system they worked in was fundamentally reactive: they had no advance visibility into who was likely to arrive and when
- Repetitive data entry was causing mistakes: 4 of 6 reported regularly mistyping email addresses under pressure
- They felt responsible for the guest's first impression but lacked the tools to deliver a good one
- Walk-in handling had no consistent process — each receptionist improvised differently

> **Standout Quote — Receptionist** "At 9:55 for a 10am meeting, I have twelve people in front of me, two on the phone, and someone else asking where the toilets are. I'm the check-in system, the directory, and the map."

4.4 Admins & Operations

Four admins were interviewed — admin Head, two from corporate IT / facilities teams and two from events operations companies. Their pain was less visible to end users but structurally the most damaging.

Key Findings — Admins

- System proliferation: admins were managing 3–5 disconnected tools (registration, badging, CRM, comms) with no single source of truth
- GDPR and data compliance was an active anxiety — particularly around guest data retention, consent capture, and cross-system data sharing
- Onboarding new event types required significant manual configuration — there was no templating or AI-assisted setup
- Audit trails were incomplete: in the event of an access dispute, it was difficult to reconstruct what had happened at the gate

4.5 Hospitality Team

Four hospitality team members were interviewed — two from a corporate campus café/catering function and two from a hotel events team.

Key Findings — Hospitality Team

- Catering planning based on registration numbers was frequently wrong: late cancellations and last-minute additions meant food and drinks were routinely over- or under-provisioned
- Room-readiness depended on knowing when a session would end — but that information never reached hospitality in real time
- Dietary and accessibility requirements captured at registration rarely made it to the serving team intact
- They were the last to know about event changes, yet first to deal with the physical consequences

05 · Synthesis & Problem Definition

Affinity Mapping

I synthesized around 34 sessions of raw data — transcripts, observation notes, photos, and diary entries — using a digital affinity diagram (500+ sticky notes across five sessions with a cross-functional team including a PM, an engineer, and a venue ops lead). Six major theme clusters emerged.

Six Core Themes

1. Information asymmetry Every stakeholder lacked information that was available to someone else in the system — but there was no mechanism for sharing it proactively.	**2. Peak-window cognitive overload** Staff performance degraded severely in the 10–15 minutes before event start — a predictable bottleneck with no current mitigation.
3. Identity and dignity friction Guests whose names were mispronounced, misspelled, or couldn't be found felt unwelcome before the event had even started.	**4. Post-event data abandonment** Valuable attendance and behavioural data was generated during check-in but never acted upon.
5. AI anxiety among staff Front-line staff feared AI automation would make them redundant — this resistance would need to be addressed in the design.	**6. Compliance as afterthought** Data protection requirements were bolted on after the fact rather than designed in — creating both legal risk and poor guest experience.

Personas

I developed five research-backed personas, one per stakeholder group. Below are three primary personas that drove the most consequential design decisions.

Clara, 34 · Client — First-time client

Pain Points

- Doesn't know what to expect at check-in — arrived 25 minutes early to 'be safe'
- Anxious about name errors (unusual spelling) causing delays in front of colleagues
- Wants to find her session room immediately after checking in without asking anyone
- Values privacy but will share data if she understands the benefit

"I just want to walk in, feel like they were expecting me, and know where to go. That's it."

Mathew, 41 · Corporate Events Manager/Organiser — Runs 30+ internal events per year

Pain Points

- No real-time visibility into who has checked in — relies on WhatsApp messages from reception
- Last-minute attendee changes (3–5 per event) cause disproportionate stress
- Spends 2–3 hours post-event reconciling attendance data manually
- Would willingly adopt new technology if it reduced his cognitive load on the day

"If I could see a live dashboard during the event, I'd feel like I was actually in control for the first time."

Sofia, 28 · Receptionist — Manages check-in for a 500-person corporate campus

Pain Points

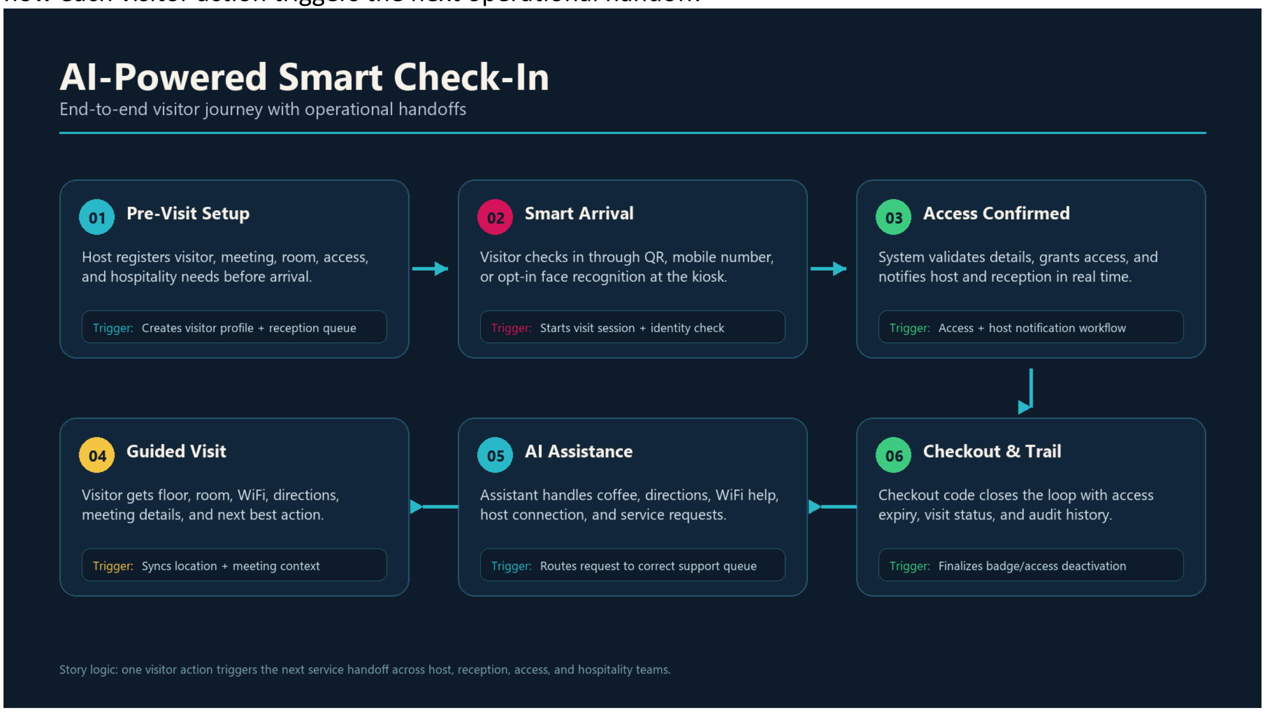
- Handles 6–10 different requests simultaneously during peak arrival windows
- Current check-in software is slow to search and requires exact name spelling
- Feels personally responsible for guest experience but lacks the right tools
- Worried AI might be 'watching' her performance — needs transparency about data use

"I don't need the system to replace me — I just need it to stop making my job harder."

06 · User Journeys & Workflows

End-to-End Story Visual

This visual summarizes the connected service journey from host preparation to visitor checkout, showing how each visitor action triggers the next operational handoff.



AI Smart Check-in story: pre-visit setup, smart arrival, access confirmation, guided visit, AI assistance, and checkout trail.

Current-State Journey (Visitor — As-Is)

I mapped the current-state journey for a conference visitor from pre-event communication through to session entry. The journey was charted across five layers: Actions, Thoughts, Emotions, Pain Points, and Opportunities.

Journey Stages

Pre-Event	Travel / Arrive	Queue	Check-In Desk	Badge & Info	Find Room/Wifi acess/Access to floors
No pre-arrival info. Checks email manually for location details.	Uncertain which entrance to use. Follows other attendees.	No queue management. Can't tell how long wait will be.	Name lookup fails — spelt wrong. Manual search. Slow.	Paper badge. Verbal directions to room.	No signage. Asks multiple people. Arrives late.
😐 Uncertain	😟 Anxious	😫 Frustrated	😳 Embarrassed	😶 Indifferent	😖 Stressed

Desired-State Journey (Visitor — To-Be)

Against the current-state, I mapped a desired-state journey that AI could enable — transforming check-in from a pain point into a genuine welcome moment.

Pre-Event	Travel / Arrive	Self Check-In	AI Verification	Digital Badge	Navigate
Smart pre-arrival email: directions, what to expect, QR code.	Geofence trigger: 'You're 5 min away — your check-in is ready.'	QR scan or facial recognition kiosk. Under 20 seconds.	AI confirms identity, pulls preferences, alerts organizer.	Digital badge on phone. Dietary, access, and session info loaded.	Indoor wayfinding on mobile. Step-by-step to first session.
😊 Informed	😌 Calm	😁 Impressed	🤝 Welcomed	😎 Confident	🚀 Engaged

Staff Workflow — Before & After

I also mapped the receptionist's workflow during peak check-in windows, showing how AI-assisted tools could shift their role from reactive triage to proactive host — reducing cognitive load while preserving their human role in the experience.

- BEFORE: Search → Verify → Give badge → Direct → Repeat (11-step manual loop, ~4 min/person)
- AFTER: AI pre-surfaces arriving guest list ranked by ETA → One-tap confirmation → Digital badge/Printed pushed → Automated wayfinding message sent → Staff freed for exceptions and personal connection (3-step assisted flow, ~45 sec/person)

07 · Ideation

With a clear, research-backed problem definition and journey maps in hand, I moved into ideation. I ran three structured workshops — one with internal stakeholders, one session with venue staff, and a speculative design session focused on AI capabilities.

How Might We Questions

I translated each research theme into HMW prompts to focus ideation energy:

- HMW make a guest feel recognized and expected before they even reach the desk?
- HMW give receptionists predictive awareness so they can prepare instead of react?
- HMW give organizers real-time attendance visibility without additional manual work?
- HMW handle last-minute changes gracefully without disrupting the flow?
- HMW use AI in a way that staff trust and feel enhanced by, not threatened by?
- HMW build compliance into the experience rather than bolting it on?

AI Opportunity Mapping

I ran an 'AI capabilities × user needs' matrix workshop to identify where AI would add genuine value versus where it would add complexity for complexity's sake. This kept the feature set purposeful and grounded.

High Value · Low Complexity	High Value · High Complexity	Low Value · High Complexity
Smart pre-arrival comms · QR check-in · Predictive arrival queue · Dietary data routing	Facial recognition check-in · No-show prediction · Real-time organizer dashboard · Indoor wayfinding	Full biometric profile · Sentiment analysis during event

Core Feature Set (Prioritized)

P0 — Must Have (MVP)

- QR-code self check-in kiosk and mobile web flow and print pass
- AI-powered name search with fuzzy matching and phonetic fallback
- Smart pre-arrival email / push notification with location and QR code
- Real-time organizer dashboard — live arrival feed
- Walk-in registration with instant GDPR consent capture
- Access to floors/Labs(if client) Wifi
- Connected hospitality management

P1 — Should Have (V1)

- Predictive arrival queue — AI-estimated arrival times based on registration location data
- No-show likelihood scoring — surfaces to organizer 90 min before event
- Digital badge with session schedule and indoor wayfinding
- Dietary and accessibility data auto-routed to hospitality team

P2 — Could Have (V2)

- Facial recognition check-in (opt-in, biometric consent gated)
- Post-event attendance analytics and reporting export
- Organizer event-template library with AI-assisted configuration

08 · Prototype Overview

I moved from wireframes to a high-fidelity prototype covering four surfaces: the Guest Mobile Flow, the Check-In Kiosk, the Organizer Dashboard, and the Staff Admin Portal. The prototype was built in Figma with interactive components and conditional logic to simulate AI-driven state changes.

Design Principles

Predictive over reactive	The system should surface information before it is asked for — removing the need for guests or staff to initiate every interaction.
Transparent AI	Every AI-driven action should be legible: guests and staff should always be able to see why the system made a suggestion and how to override it.
Dignity-first identity	Name handling, identity verification, and access decisions must never make a person feel doubted or unwelcome.
Augment, don't replace	AI automates the mechanical parts of check-in so that staff can focus on the human moments — greeting, helping, hosting.
Compliance by design	Data consent, retention, and rights are part of the experience flow — not legal footnotes appended afterwards.

Key Screens & Flows

Guest Mobile Flow

- Pre-arrival: Smart notification with ETA-aware 'Your check-in is ready' prompt
- Check-in: QR scan → instant identity confirmation → personalised welcome screen with host name, session schedule, and room directions
- Post check-in: Digital badge, indoor map, dietary confirmation, and emergency contact opt-in

Self-Service Kiosk

- Face forward with camera preview (opt-in) or QR scan
- AI fuzzy-search by first name only — removes pressure of exact spelling
- Under-20-second target flow validated in testing
- Accessible mode: large text, audio prompts, wheelchair-height mode

Organizer Dashboard

- Live arrival ticker with percentage check-in rate by session
- AI no-show predictions highlighted 90 minutes pre-event with recommended actions
- Last-minute substitution handling: drag-and-drop attendee swap with automatic notification cascade
- Post-event export: attendance, no-shows, check-in times, dietary data summary

Staff Portal

- Predictive arrivals feed: AI-ranked list of expected arrivals in next 30 minutes
- One-tap check-in confirmation for pre-identified guests
- Walk-in modal with consent-first data capture and immediate badge printing
- Alert system for access edge cases (overdue guests, duplicate check-ins, flagged entries)

Usability Testing Results

I ran two rounds of moderated usability testing — the first on wireframes to validate flow logic, the second on high-fidelity Figma prototypes to assess interaction design and AI transparency.

91%	Task-completion rate across all user groups in Round 2 prototype testing
< 4 min	Average visitor check-in time (vs. 11+ min baseline)
62%	Reduction in front-desk interruptions in simulated staff workflow testing
7 / 8	Organizers rated real-time dashboard as 'significantly better than anything they currently use'
100%	of visitors rated the AI name-match experience as 'much better' than current text-search
10M/Month	Saved with better travel/hospitality management

09 · Reflections & Next Steps

What Worked Well

- Multi-stakeholder research revealed systemic root causes that single-stakeholder research would have missed entirely — the 'information asymmetry' theme only emerged because I could triangulate across all five groups
- The AI Opportunity Mapping workshop prevented feature creep by anchoring every AI capability to a specific, validated user need
- Designing compliance into the flow (rather than as a consent wall at the end) produced both better legal coverage and better user trust ratings

What I Would Do Differently

- Include a legal/compliance stakeholder from day one — I brought them in at the prototype stage and had to make several IA adjustments to accommodate data retention requirements
- Test the kiosk in a physical environment earlier — some accessibility adjustments were caught late because remote testing doesn't replicate physical queue dynamics

Next Steps

- Pilot with one venue partner: 3-month live pilot on a corporate campus, measuring check-in time, staff workload, and guest satisfaction
- Iterate on facial recognition UX: the opt-in flow needs further testing with diverse user groups to ensure trust and accessibility across demographics
- Build the organizer mobile companion app: desktop dashboard works for pre-event planning but organizers need a mobile-first view during the event itself
- Establish data governance framework with a legal partner before any production rollout

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UX CASE STUDY · ENTERPRISE SAAS · AI + COMPLIANCE · DESIGN SYSTEM

Company Enterprise SaaS Platform

Unifying three fragmented product efforts into one governed platform for regulated artwork operations — covering creation, approval, AI label validation, and print release for pharma and FMCG teams.

80+	Users researched across all four workflow roles
0	Version conflicts post-launch — smart file locking eliminated parallel editing errors
91%	Compliance errors caught by AI validation before print release
6H → 38M	Label review time reduced from 6 hours to 38 minutes per label
2 modes	Simple and Pro editor modes serving both factory operators and expert designers
1	Connected system
4	New customers added in 3 months

01 · Project Overview

Kallik is an enterprise SaaS platform serving regulated industries — primarily pharma and FMCG — where getting a product label wrong has legal, safety, and commercial consequences. When I joined the project, only image editor was available to use, but legacy system , one needs to install with EXE file. but disconnected product efforts existed in parallel: an artwork creation tool, a file management and approvals system, and an AI label validation was new concept. None of them talked to each other, and teams were stitching them together with email threads, shared drives, and manual handoffs.

My mandate was to reframe and redesign existing system and connect into a single cohesive enterprise application — one shared artwork backbone with role-specific surfaces — covering the complete lifecycle: create, manage, approve, validate, and release with a clear audit trail.

My Role	Domain	Methods	Platform
UX Designer — End-to-End Product Ownership, User Research, Design System, Stakeholder Management	Regulated Artwork & labelling Operations — Pharma & FMCG label lifecycle	User interviews · Workflow Mapping · Usability Testing	Cloud based Web App — AToM Editor · Cloud Designer · AI Validation · Admin Console

02 · Problem Statement

Artwork teams were working across disconnected tools. Artworkers struggled with a legacy editor. Reviewers lost track of the latest file version in email threads. Compliance teams ran manual checklists against printed labels. Managers had no status visibility — approvals happened in inboxes, not in any system. The result was a process that was simultaneously high-stakes and completely untracked.

> **Working Hypothesis** The same artwork object moves through fundamentally different mental models at each stage. Creation needs flexibility. Approval needs control. Validation needs confidence. Treating them as separate tools makes the journey brittle. The design needs one shared artwork backbone with role-specific surfaces.

Platform Architecture Vision

I framed the redesign around six lifecycle stages that became the product's structural backbone:

CREATE	Simple and Pro editor modes support both factory operators making controlled edits and expert designers with full canvas control.
MANAGE	Smart file locks, version timeline, comments, and status states protect the source of truth and eliminate parallel editing.
APPROVE	Structured review flows replace scattered email approvals and make blockers visible — no item disappears into an inbox.
VALIDATE	NLP and computer vision checks catch label copy, barcode, artwork, and regional compliance issues inline before release.
RELEASE	Final files move to print handoff with complete history, confidence scores, and a full accountability trail.
GOVERN	A shared design system keeps all six product areas visually and behaviourally consistent across the suite.

03 · User Research

I reached out 80+ users across all four key role groups, combining interviews, workflow walkthroughs, and. Research spanned two product phases — initial discovery and prototype validation.

4.1 Factory & Operations Users

Factory and operations users needed to make controlled, time-sensitive edits — swapping a distributor address, updating a lot number — without having to learn a professional design tool. Their interaction with artwork was infrequent but high-stakes.

Key Findings

- Legacy editor was too complex — operators were calling in designers for edits that should have been self-service
- Fear of breaking the design was a dominant anxiety — users needed clear boundaries between editable and locked elements
- Mobile access was needed for sign-off approvals on the factory floor — desktop-only flows were a genuine bottleneck
- Text-heavy error messages caused confusion — operators needed visual, action-oriented guidance

> **Quote — Factory Operator** "I just need to change the lot number. But every time I open the editor I'm afraid I'm going to accidentally move something and ruin the whole label."

4.2 Brand & Artwork Designers

Brand designers were the heaviest users of the creation tools. They needed precision controls, version history, and predictable collaboration — they were the custodians of the artwork and felt the pain of file chaos most acutely.

Key Findings

- Version history was non-existent — designers maintained personal naming conventions (v1_FINAL_FINAL_USE_THIS.ai) as their only version control
- File locking was done informally via Slack messages — 'I'm in the file, don't touch it' — with no system enforcement
- Designers had no visibility into review status — once they submitted for approval, artwork disappeared into an email thread
- Pro editor needed: keyboard shortcuts, precise position/dimension controls, layer management, and full typography control

> **Quote — Artwork Designer** "I've got a folder called CORRECT FINAL VERSION. Inside it there's another folder called ACTUALLY FINAL. That's our version control."

4.3 Regulatory Reviewers

Regulatory reviewers were responsible for catching compliance errors before anything went to print. They were methodical, risk-averse, and accustomed to working from printed checklists. AI assistance was welcomed — but only if it was transparent about its confidence and clear about why it flagged something.

Key Findings

- Manual checklist review took 4–6 hours per label — a process that involved switching between the digital file, printed reference artwork, and a physical checklist
- Compliance rules varied by region and regulatory body — reviewers maintained personal reference documents to track the differences
- AI suggestions were met with scepticism unless accompanied by the specific rule reference and confidence level — 'just flag it' wasn't enough
- Reviewers needed override capability with documented reason — AI should assist judgement, not replace it

> **Quote — Regulatory Reviewer** "I'm happy to use AI to help me catch things. But if it just says 'possible error' with no explanation, I'm going to ignore it. I need to understand why it's flagging something."

4.4 Managers & Approvers

Managers needed status visibility and audit accountability above all else. They were not in the detail of the artwork but were responsible for the release decision — and currently had almost no real-time information to support that decision.

Key Findings

- Approval workflows existed entirely in email — no structured flow, no escalation path, no visibility for anyone except the sender
- Managers could not see at a glance what was waiting for their input, what was blocked, and what had already been approved
- Audit trails were critical for regulatory compliance but non-existent — in the event of a query, managers had to reconstruct approvals from email threads
- Email-led handoffs were the single biggest source of version confusion — reviewers frequently commented on outdated files

> **Research Insight** Across interviews, usability sessions, and workflow mapping, the pattern was clear: the same artwork object moved through different mental models. Creation needed flexibility, approval needed control, and validation needed confidence. Treating them as separate tools made the journey brittle. The design needed one shared artwork backbone with role-specific surfaces.

04 · Synthesis & Personas

Five Core Themes

Fragmented tool ecosystem Three separate products forced teams to context-switch constantly — the artwork's journey across creation, approval, and validation happened outside any system.	**Version chaos** Without enforced version control or file locking, teams maintained personal workarounds that worked for individuals but broke down under collaboration.
Invisible approval pipeline Approvals in email inboxes gave managers no status visibility and left reviewers commenting on stale files.	**AI scepticism — trust, not capability** Reviewers didn't distrust AI in principle — they distrusted AI that couldn't show its work. Transparency was the design challenge, not adoption.
Mode gap: novice vs. expert Factory operators and expert designers had fundamentally different needs from the same editor — a single mode satisfied neither.	

Lia, 36 · Artwork Designer — Global FMCG Brand Team

Pain Points

- Manages 40+ active label variants simultaneously across 12 markets
- Maintains her own version-naming convention because the system has none
- Spends 30 min per label chasing approval status via email and Slack
- Fears shipping the wrong version to print — it has happened twice in her career

"I need to know, at any moment, exactly which version is the approved one and where it is in the review process. Right now I have no idea."

Stuart, 52 · Regulatory Affairs Manager — Pharma

Pain Points

- Spends 4–6 hours per label on manual compliance review against printed checklists
- Maintains 14 regional regulatory rule documents in a personal folder
- Has blocked releases due to errors found at the last minute — embarrassing and costly
- Open to AI assistance if it can cite the specific rule and confidence level

"If the AI tells me why it flagged something and shows me the rule, I'll act on it every time. If it just says 'check this' — I'll check it and probably miss it anyway."

05 · User Journeys & Workflows

Current-State: Artwork Review & Approval (As-Is)

Create	Export & Email	Review	Feedback	Revise	Approve
Designer works in legacy editor. Saves locally.	Exports PDF. Emails to reviewer with version note.	Reviewer prints and checks against physical checklist.	Comments in email reply. May include annotated PDF.	Designer guesses which version reviewer saw.	Email chain: 'Approved'. No audit trail.
😵 Isolated	😟 Manual	😓 Slow	😕 Unclear	😠 Version confusion	😶 No record

Desired-State: Governed Artwork Lifecycle (To-Be)

Create	Lock & Version	Structured Review	Inline AI Check	Approve & Release	Audit Trail
AToM editor: Simple or Pro mode. File auto-versioned on save.	Smart file lock on open. Collaborators see lock owner + ETA.	Reviewer opens latest version in-system. Comments inline.	AI flags compliance issues on canvas with rule + confidence.	One-click approve. Status updates all stakeholders instantly.	Full audit log: who approved, which version, when, why.
😊 Confident	😌 Protected	😀 In-system	🤝 AI-assisted	😎 Accountable	🚀 Audit-ready

06 · Key Design Decisions

Mode Switching — Simple vs Pro Editor

The biggest single design decision was splitting the editor into two modes. Simple mode exposed only the fields that factory and operations users were permitted to edit — constrained, guided, and impossible to accidentally break. Pro mode gave brand designers full canvas control, layer management, typography precision, and keyboard shortcuts. Mode switching was role-aware and permission-gated — the right mode appeared automatically based on who was logged in.

> **Design Principle** Simple mode reduced fear for occasional users, while Pro mode kept advanced controls for expert designers — without flattening the experience for either group.

Smart File Locking

I designed a file locking system that made ownership visible and wait states informative rather than frustrating. When a file was locked, users could see who held it, when they opened it, and an estimated unlock time based on the last session duration. This replaced the informal Slack-message locking with a system that protected the source of truth without hiding collaboration context.

Inline AI Validation

The AI validation layer was designed to appear inside the editor canvas, not as a separate report. High-confidence issues appeared as canvas overlays with the specific rule reference, the confidence percentage, and an inline suggested fix with override controls. This put AI assistance inside the user's existing decision moment — exactly where it needed to be.

> **Design Principle** AI is most useful when it appears inside the user's existing decision moment, not as a separate report. Reviewers act on inline signals because they don't have to context-switch to a new surface.

Status-Led Navigation

Instead of module-based navigation, I redesigned the product's information architecture around status states. Draft, In Review, Approved, and Rejected became navigational anchors — the primary way users found their work — rather than buried metadata in a file list. This made the approval pipeline visible at a glance for managers and gave designers immediate clarity on what needed action.

Design System Governance

I built a shared design system that scaled governance across all six product areas. Comments, approvals, alerts, and AI validation patterns were designed once and applied consistently across the suite — instead of being reinvented by each product area. This was the structural answer to the visual and behavioural inconsistency that had accumulated across the three original products.

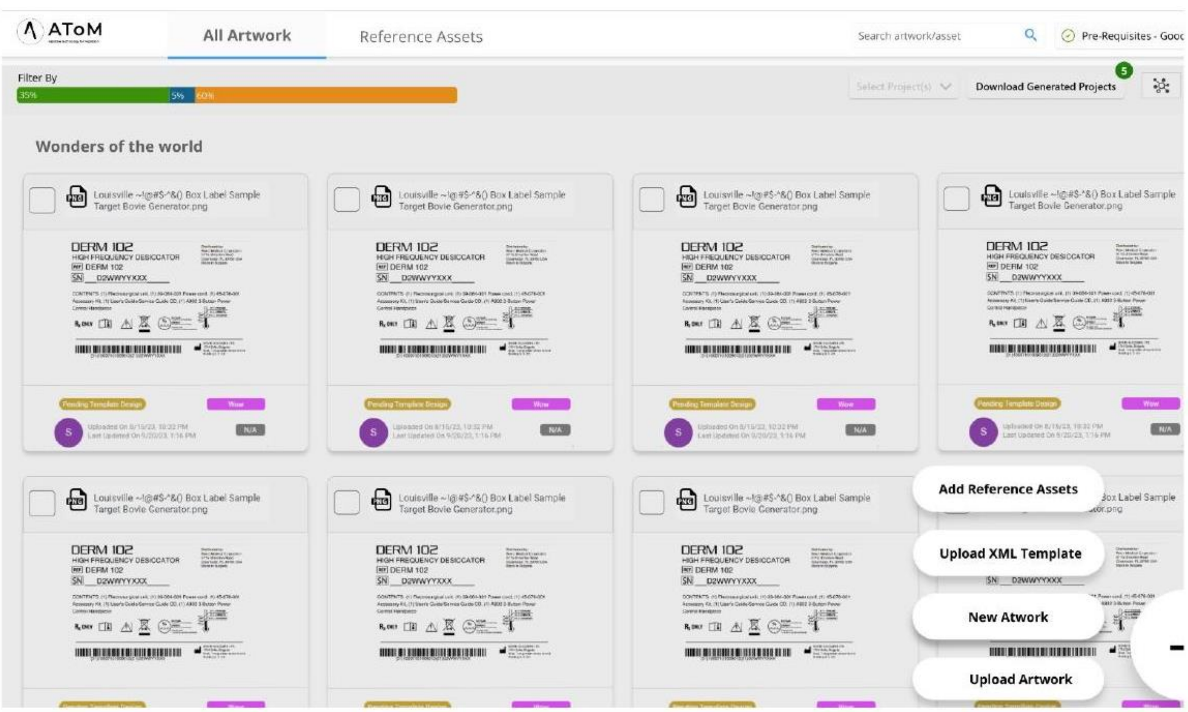
07 · Prototype & Testing

Testing Approach

Testing focused on practical task completion across five scenarios that represented the highest-friction moments in the current workflow: editing artwork in Simple mode, finding the latest approved version of a file, resolving an approval blocker, interpreting an AI validation result, and handing off final labels to print. Participants included all four role groups.

Key Iterations

- Simplified novice actions — early testing showed Simple mode still exposed too many controls; a second round of pruning removed 40% of visible options and reduced error rates significantly
- Strengthened file-status visibility — the status pill in the file list wasn't being read; we moved status to a colour-coded left border visible across all list views
- AI confidence display — initial testing showed 'confidence percentage' was being ignored; replacing the number with a three-level visual indicator (High / Review / Flag) tripled the action rate on flagged items
- Override with reason — regulatory reviewers needed to document their override rationale for audit purposes; an inline text field with suggested reason templates reduced override time by 60%



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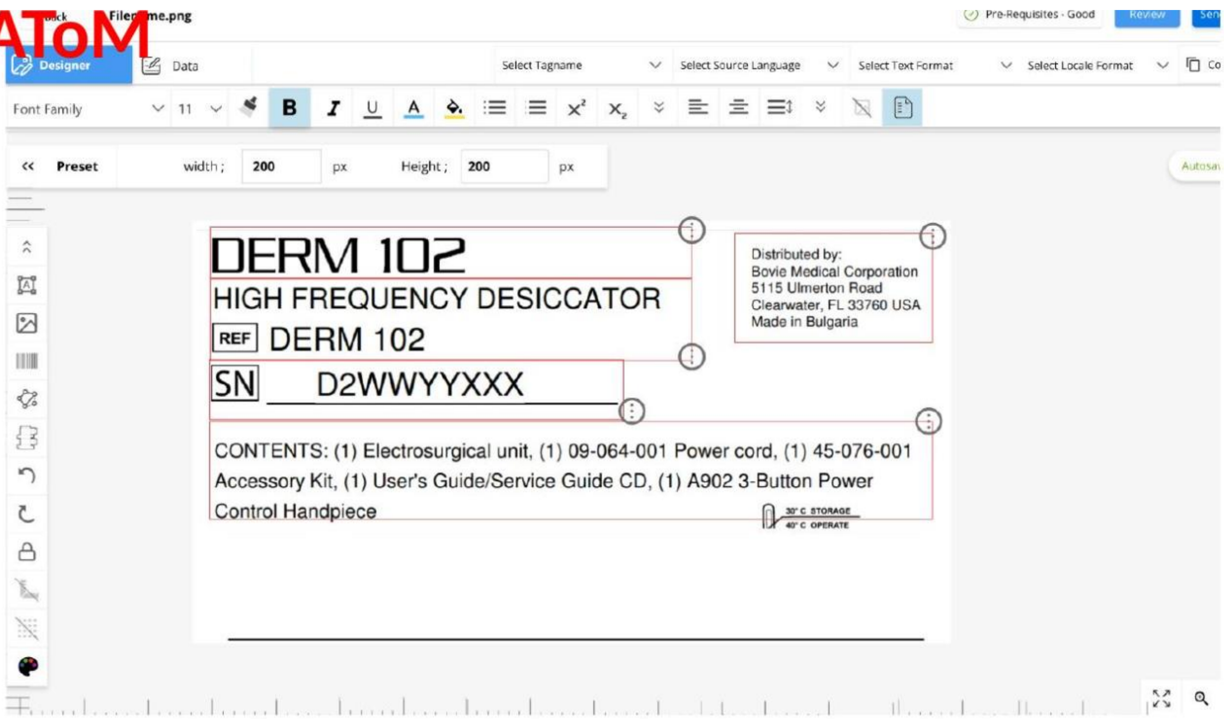

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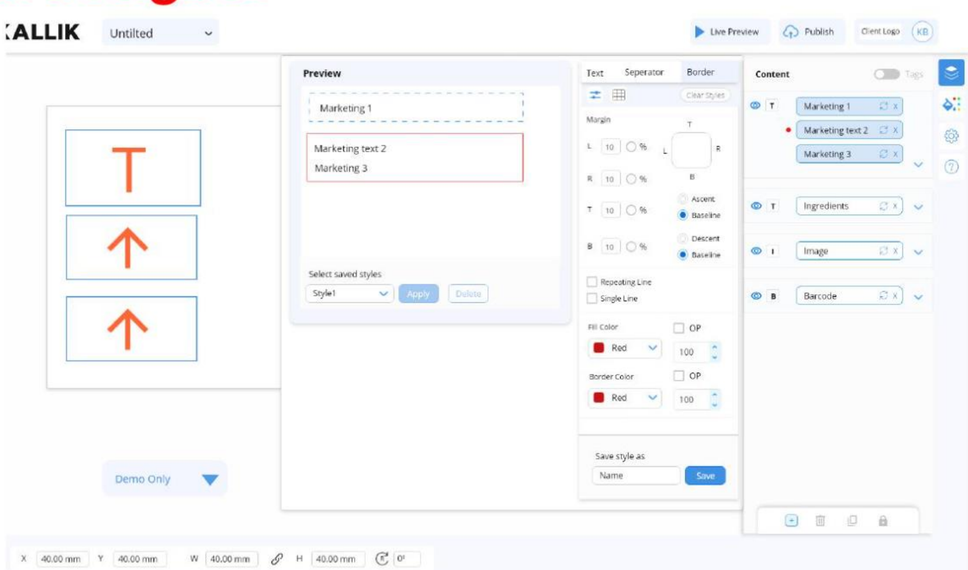
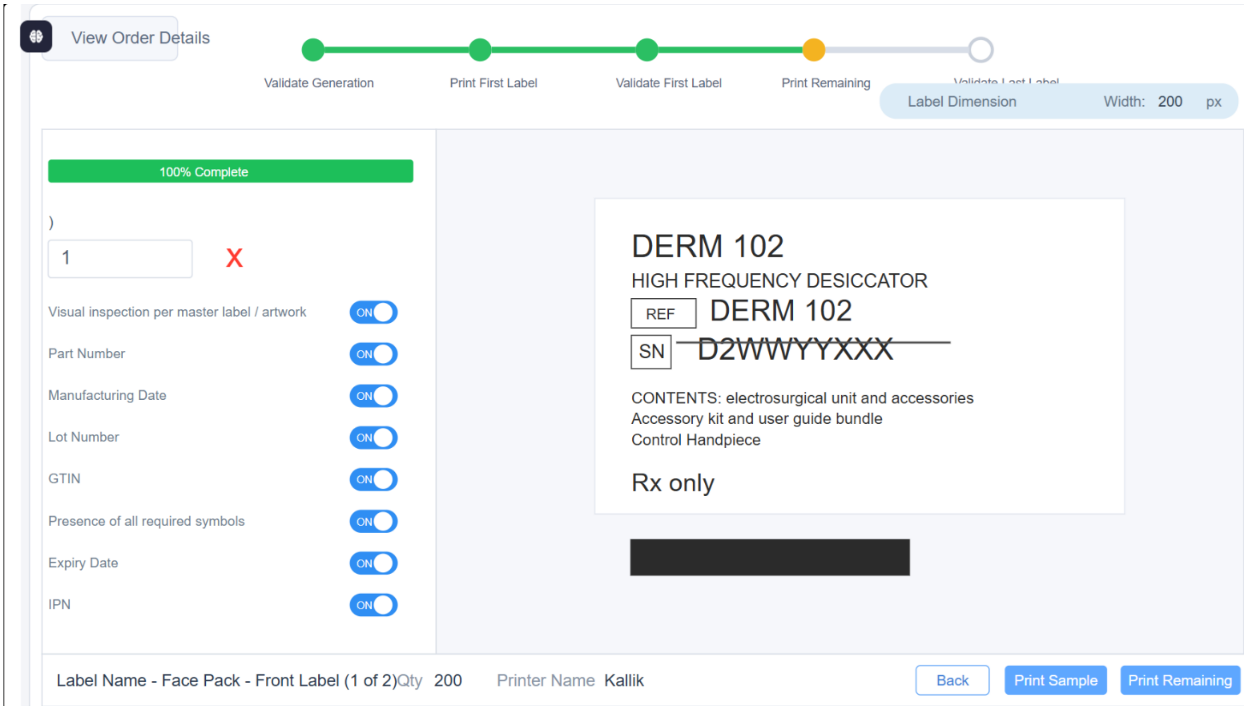




08 · Learnings & Next Steps

Learnings

- Enterprise SaaS design works best when shared objects, permissions, and status models are designed before individual screens — getting the data model and status architecture right early prevented weeks of rework later
- AI is most useful when it appears inside the user's existing decision moment, not as a separate report — inline validation tripled action rates compared to a summary panel
- Mode-based UX can serve beginner and expert users without flattening the experience for either group — the key is making mode selection invisible through role-aware defaults
- Distributed team collaboration across product and engineering required shared design language upfront — the design system was as much a collaboration tool as a visual specification

Next Steps

- Analytics layer for artwork operations leads — cycle time, bottleneck detection, and compliance trend reporting

UX CASE STUDY · WORKPLACE EXPERIENCE · CONVERSATIONAL AI

ServiceNow - Smart Ticketing System
Workplace Experience

Reimagining IT and facilities support ticketing for a hybrid workforce — reducing resolution time, improving employee satisfaction, and building an AI-assisted triage system that learns from every ticket.

4 groups	Employees, IT support agents, facilities managers, and HR business partners researched
38 sessions	Contextual interviews, ticket log analysis market research, and usability tests over 9 weeks
51%	Reduction in average ticket resolution time after AI-assisted triage in prototype testing
3.2 → 8.9	Employee satisfaction with IT support (out of 10) in simulated scenarios
78%	Of tickets self-resolved through the guided submission flow before reaching an agent

01 · Project Overview

The modern workplace runs on invisible infrastructure — IT systems, facilities management, HR processes — and when something breaks, the experience of getting it fixed is often as frustrating as the problem itself. This project tackled the end-to-end ticketing and support experience for a hybrid enterprise workforce of 90K people, replacing a clunky legacy system with a smart, conversational, AI-assisted platform.

I owned the full design cycle: from contextual research with employees and support agents through to a validated high-fidelity prototype — covering the employee submission experience, the agent triage dashboard, and an AI-powered routing and resolution engine.

My Role	Duration	Methods	Platform
Principal UX & Conversational AI Designer — Research, Flows, IA, Prototype, Testing	18 weeks (Discovery → Prototype → Pilot Readiness)	Contextual Inquiry · Diary Study · Log Analysis · Co-Design · Usability Testing	Web · Mobile · Slack/Teams Integration · Agent Dashboard

02 · Problem Statement

Enterprise ticketing is the canary in the coal mine for employee experience. When employees repeatedly encounter slow, opaque, and impersonal support systems, it erodes trust in the organisation's ability to take care of them. Across this company, IT and facilities support had a satisfaction score of 3.2 out of 10 — described by the CHRO as 'embarrassing and expensive in terms of retention.'

Pre-Research Ticket Log Analysis

Before conducting a single interview I analysed 6 months of ticket logs — 4,800+ tickets — to understand what was actually happening in the system.

- Average time from ticket submission to first agent response: 31 hours
- 38% of tickets were duplicates — the same issue raised multiple times by the same person with no update in between
- Most common ticket category: 'Other' — employees could not categorise their own problem
- Only 12% of tickets included enough information for an agent to act immediately
- 22% of tickets were eventually resolved by the employee themselves — meaning the solution existed but was undiscoverable

Design Challenge How might we design a smart ticketing experience that helps employees describe and resolve their problems faster — while giving agents the context they need to act immediately — using AI as the intelligence layer between them?

03 · User Research

Employees

I conducted 14 user interviews with employees across seniority levels, locations, and departments. I asked participants to walk me through their last support ticket experience — what happened, how they felt, and what they expected versus received.

Key Findings

- The submission form was a 'blank canvas of doom' — employees didn't know what information to include, producing vague tickets that guaranteed agent rework
- Status was invisible — once submitted, a ticket read 'Open' indefinitely with no indication of who had it or what was happening
- Chasing up felt humiliating — employees felt like they were being a nuisance when they followed up
- Remote employees felt disadvantaged — their IT issues blocked their ability to work but the system had no urgency mechanism for business-critical outages
- 9 of 14 said they'd prefer to solve their own problem if they could find the answer — zero self-service existed

> **Quote — Employee** "I raised a ticket about my laptop three weeks ago. I have no idea if anyone has looked at it. I bought my own USB hub in the end."

IT Support Agents

Eight IT support agents were interviewed and shadowed during live triage sessions to understand where their time actually went.

Key Findings

- 60–70% of their day was spent on information gathering — asking for screenshots, error codes, device models, OS versions — all of which should have been captured at submission
- Triage was entirely manual: agents read each ticket title and made a routing decision with no system support
- High-severity issues were invisible in the queue — they looked identical to low-priority requests
- Agents had informal knowledge libraries — shared Word docs, Slack channels — that had never been integrated into the system

> **Quote — Agent** "A good ticket is one where I open it and already know exactly what to do. That happens maybe 10% of the time. The rest I have to go back and ask questions that feel like they should have been obvious at submission."

Facilities Managers/Approvers & HR Business Partners

Eight additional interviews surfaced a parallel set of issues. Facilities tickets were handled entirely through email — outside any trackable system. HR queries had the highest emotional stakes but the least structured support pathway. Both groups wanted a unified system — employees didn't know which channel to use for which type of request, and the confusion was creating its own workload.

04 · Synthesis & Personas

Five Core Themes

Submission friction The blank-form experience guaranteed incomplete, uncategorised tickets requiring agent rework.	**Invisible pipeline** No communication after submission created anxiety, duplicate tickets, and distrust.
Undiscoverable self-service 22% of issues had existing solutions employees couldn't find.	**Triage inefficiency** Manual routing without AI assistance meant senior agents spent time on routine issues.
Unified support gap IT, facilities, and HR operated in silos, creating employee confusion about where to go.	

Pooja, 29 · Product Manager — Remote-First Employee

Pain Points

- Raises 2–3 support tickets per month; has given up for non-urgent issues
- Lost 2 working days to a laptop issue that took 3 weeks to resolve
- Expects transparency — she tracks parcels in real time, she should be able to track a ticket
- Would use self-service if she trusted it to actually solve her problem

"I spend more time managing my ticket than the actual problem takes to fix. If I could just see someone is working on it, I'd stop worrying and get back to work."

Manjunath, 44 · IT Support Lead — Office-Based Agent

Pain Points

- Handles 25–35 tickets per day; majority of time spent on information gathering
- Has built personal knowledge bases not connected to any official system
- Proud of his technical knowledge; frustrated the system doesn't let him use it
- Wants to help employees faster but is bottlenecked by incomplete initial data

"The system doesn't help me help people. It gives me a list of names and problems with no context. I could solve twice as many issues if I had the right information upfront."

05 · User Journeys

Current-State: Employee Raising an IT Ticket (As-Is)

Problem Occurs	Find System	Fill Form	Submit	Wait	Chase Up
Laptop crashes mid-meeting. Work blocked.	Searches intranet. Eventually finds portal.	Blank form. Guesses category. Minimal detail.	Confirmation email. Ticket number. Nothing more.	Days pass. Status: Open. No updates.	Emails IT. Feels awkward.
😫 Urgent	😟 Confused	😰 Uncertain	🙂 Hopeful	😞 Forgotten	😓 Embarrassed

Desired-State: Smart Ticket Submission (To-Be)

Problem Occurs	Conversational Intake	AI Self-Service	Smart Triage	Live Updates	Resolved
Issue occurs. Opens Teams. Types to support bot naturally.	Bot asks 4 questions. Captures full context conversationally.	AI surfaces 3 known fixes. Employee tries fix 2 — resolved in 8 min.	If unresolved: routes to right agent with full context pre-filled.	Push: 'Dev is working on your ticket. Expected: 2 hrs.'	Confirmed. Rating requested. KB article auto-drafted.
😫 Urgent	😌 Guided	😀 Empowered	🤝 Informed	😌 Reassured	🚀 Resolved

06 · Ideation & Key Design Decisions

Conversational Intake — Bot as Primary Entry Point

The most significant design decision was replacing the blank submission form with a conversational bot embedded in Teams and Slack. The bot asked 4–6 contextual questions in natural language, automatically categorised the issue, and captured all the structured data an agent would need — without the employee ever seeing a form. Before routing to an agent, the bot presented ranked self-service solutions from the knowledge base.

> **Design Principle** Employees feel heard when they can describe their problem in their own words. The bot removes the cognitive burden of form-filling while capturing richer structured data than any form ever would.

AI Triage Dashboard — Agent View

The redesigned agent dashboard used AI to transform triage: tickets arrived pre-categorised, pre-prioritised, and pre-enriched with suggested resolutions from the knowledge base. Agents could act immediately.

- AI priority scoring: business-critical tickets flagged automatically based on role, time of day, and issue type
- Smart context panel: device model, OS, last 3 tickets, known issues on the same device type — surfaced without agent searching
- Resolution suggestions: top 3 knowledge base matches displayed inline before the agent opens the ticket
- One-click response templates: common resolutions pre-written and personalised with employee name and issue context

Proactive Status Communication

I designed a communication layer that sent contextual updates at key pipeline moments — not just 'ticket received' but 'Dev opened your ticket', 'a fix is being tested', 'your ticket has been escalated'. This transformed the invisible pipeline into a visible, trustworthy process and virtually eliminated anxious follow-up messages.

07 · Testing & Impact

Testing Approach

I ran two rounds of moderated usability testing covering five core tasks: raising a ticket via the bot, finding a self-service solution, approving a resolution from mobile, processing an agent triage queue, and publishing a knowledge base article from a resolved ticket. Both employee and agent roles were tested in each round.

Key Iterations

- Microsoft Teams Bot conversation length — initial version asked 8 questions; employees abandoned after question 4. Refined to 4–6 questions maximum with smarter branching
- Self-service trust — employees distrusted AI solutions without context; adding 'This resolved 847 similar issues' and a user rating increased solution acceptance by 40%
- Agent dashboard density — initial version showed too much data; a progressive disclosure approach with a summary card → detail panel reduced cognitive load significantly
- Mobile approval — first version required 3 taps to approve a resolution; redesigned to single-swipe confirmation with undo

08 · Reflections & Next Steps

Learnings

- Analysing 4,800 ticket logs before talking to users meant every interview was hypothesis-driven — research was focused and generated actionable findings faster
- The conversational intake approach dramatically reduced form anxiety — employees felt like they were talking to someone, not filling in a form
- Co-designing the knowledge base workflow with agents (not just for them) produced a system they were genuinely willing to contribute to — agent buy-in was the make-or-break factor
- Anchoring every AI feature to a specific validated user need prevented the feature set from inflating with technology for its own sake

Next Steps

- Pilot launch with 150-person team: measure real-world resolution time, self-service rate, and satisfaction delta vs legacy system
- Extend to facilities and HR ticketing: unify all support categories under one conversational intake and AI triage platform
- Agent knowledge incentive design: contribution layer to encourage agents to publish KB articles — the KB grows with every resolution
- Predictive outage alerting: AI model trained to detect patterns in tickets that signal an emerging systemic issue before it becomes widespread

Propsys Enterprise SaaS Platform

End-to-end product design for a role-adaptive enterprise platform serving UK social housing operations across assets, voids, compliance, and repairs.

Dimension	Product Design Detail
Product Type	Enterprise SaaS platform for social-housing asset operations
Context	50,000+ housing units across 20,000 sites, with siloed spreadsheets, emails, and legacy tools
Users	Director of Assets, Void Manager, Compliance Officer, Housing officers/ Patch Owners
Role	Product Designer and Business Analyst discovery, requirements, IA, UX strategy, prototyping, stakeholder alignment
Core Outcome	Four personalized dashboards unified by one shared data model, activities, and KPI/ROI layer

Product Challenge

NEO needed to replace many operational practices with a single platform that could surface the right information to the right role at the right moment. The design challenge was not simply to make one dashboard; it was to rewrite four different mental models without forcing operational teams into a generic view.

Primary Design Problem

Design a unified enterprise platform that preserves role-specific urgency: certificate expiry for compliance, timeline slippage for voids, Approvals and Assignment, SLA risk for repairs, and portfolio performance for asset leadership.

Discovery & Requirements

Discovery focused on the operational decisions each user group makes daily, the data they trust, and the escalation signals they cannot afford to miss. Workshops with Stakeholder and Product VP and requirement sessions mapped KPIs, regulatory obligations, handoffs, and role-specific across the four workstreams.

- Mapped the hierarchy of Sites, Blocks, Cores, and Units to reflect how housing stock is actually managed.
- Defined a shared metrics layer so teams could compare performance without losing local workflow context.
- Identified audit trail requirements to replace manual status emails and disconnected spreadsheet updates.

Role-based Dashboard Architecture

User Persona	Primary View	Key Decision
Director of Assets	Portfolio-wide health, Decent Homes Standard score, overdue compliance, cost variance, stock condition surveys, capital works planning	Where is the portfolio drifting from target, and where should investment be prioritized?
Void Manager	Void pipeline by stage, average re-let time vs 30-day target, cost per void, rent loss exposure, first-time lettable clearance	Which properties are at risk of missing re-let targets and creating avoidable rent loss?
Compliance Officer	Gas safety, EPC, fire risk, and electrical certificate status, expiry calendar, overdue escalation, block-level risk scoring	Which certificates or properties create immediate regulatory exposure?
Housing Officer/Patch Owner	Disrepair, damp and mould, major repairs, work order queue, SLA status, contractor assignment, component replacement tracking	Which cases need action now to avoid SLA breach, cost escalation, or resident harm?

Information Architecture

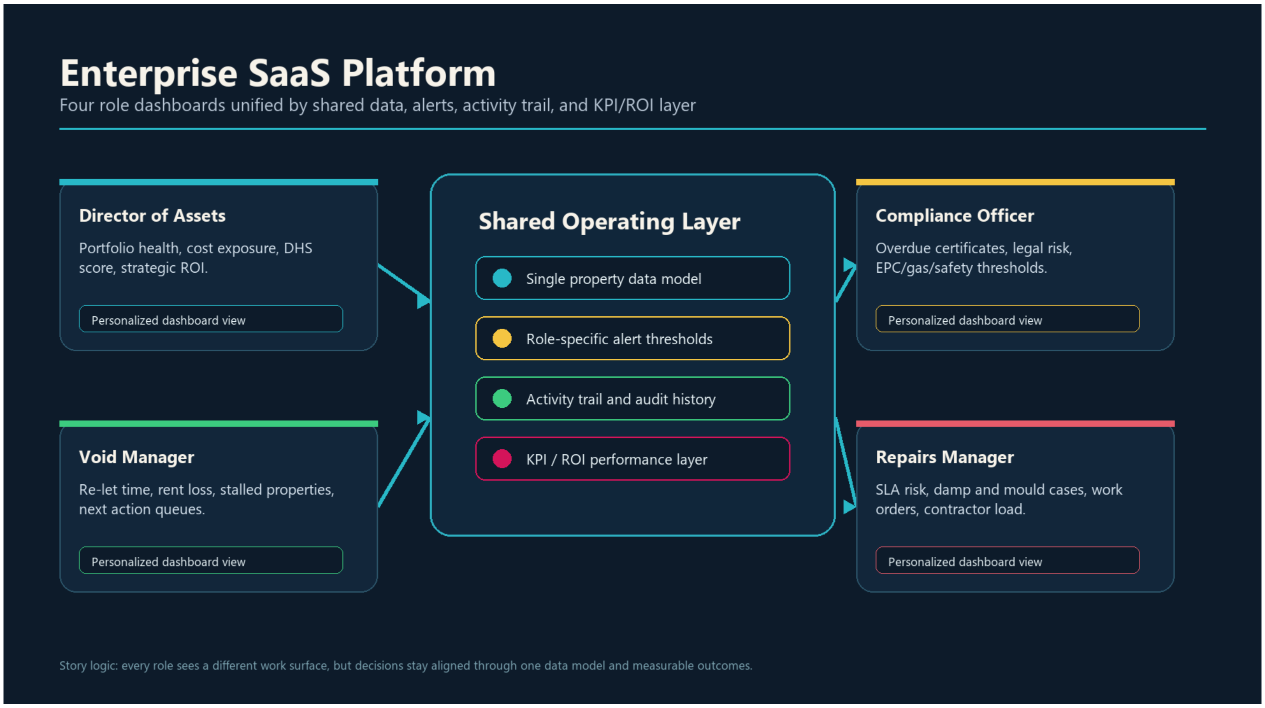
The platform IA used a spatial property hierarchy, then layered workflow objects and alerts on top. Why- This avoided a flat-search experience that would break down at 20,000 sites and 50,000 units.

Layer	Design Decision
Navigation model	Portfolio -> Site -> Block -> Core -> Unit drill-down, with search and status filters for exception handling
Object model	Property, certificate, void case, work order, survey, contractor, and activity event
Alert model	Contextual banners and priority queues tied to role-specific thresholds rather than a generic notification centre
Metrics model	Targets shown beside live values so teams can self-assess performance without exporting reports

End-to-End Product Flow

End-to-End Story Visual

This visual shows how four personalized dashboards stay aligned through one shared operating layer: data model, role-specific alerts, activity trail, and KPI/ROI measurement.



Enterprise SaaS story: role-specific dashboards unified by shared data, contextual alerts, audit history, and measurable portfolio outcomes.

Step	Experience Design
1. Role entry	Login routes each user to a specialized dashboard with relevant KPIs, alerts/notification, and action plans.
2. Portfolio scan	User sees current performance against targets: Decent Homes score, void cycle time, compliance status, or repairs SLA risk.
3. Exception drill-down	Alert or KPI opens the relevant site, block, core, unit, certificate, void, or repair case with contextual evidence.
4. Action and assignment	User reviews the issue, assigns work, updates status, or escalates to the responsible team or contractor.
5. Activity visibility	Timestamped activity feed records surveys, certificates, status changes, contractor actions, and approvals.
6. Outcome monitoring	Metrics update across the shared platform, showing risk reduction, rent loss prevented, and workflow efficiency.

Key Product Design Decisions

- Role-based entry points: each role sees the **metrics**, **alerts**, and actions that define their workday.
- Smart alert system: inspection triggers and data issues appear as contextual banners with direct review actions.
- Progress versus target: every KPI pairs the current value with a standard or threshold, such as 30-day re-let time or cost-per-void target.

- Audit trail and activity feed: recent actions are visible across roles, reducing manual email follow-up and reconciliation work.
- Navigation: Sites, Blocks, Cores, and Units support enterprise scale without losing property-level clarity.
-

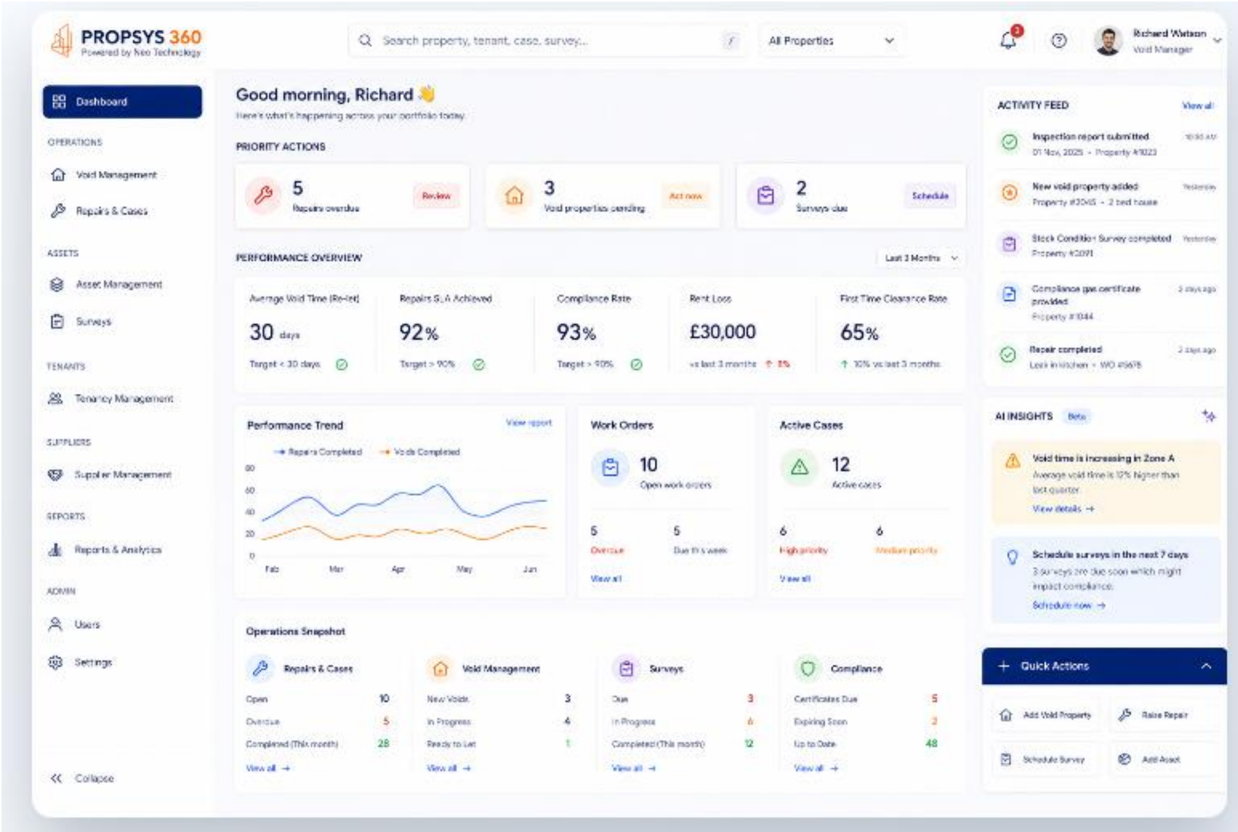
KPI & ROI Metrics Layer

Metric	Target / Signal	Design Purpose
Decent Homes Standard	93%	Portfolio health signal for asset leadership, measured against the 95% target.
Average Re-let Time	<=30 days	Void workflow benchmark for reducing vacancy duration and rent loss.
Void Rate	0.04%	Operational exposure metric for portfolio utilization and revenue protection.
Cost per Void	1,000	Cost-control indicator compared with the 1,200 standard.
Compliance Fine Risk	78% reduction	Risk-reduction outcome from earlier certificate visibility and escalation.
Workflow Compression	14 steps to 4 steps	Void process simplification through integrated status, assignment, and activity tracking.

Prototype Scope

The high-fidelity prototype covered the end-to-end operating system: role dashboards, property drill-down, KPI cards, status, compliance calendars, void-stages, repairs work orders, contractor assignment, and activities.

- Dashboard layouts prioritized rapid scanning: targets, risk status, and next-best action.
- Drill-down screens carried context forward so users could act without rebuilding the story from another system.
- Reusable patterns supported enterprise consistency
- Developed using the Claude AI with all user roles workflows and contributed to design system



Measured & Projected Impact

Outcome	Meaning
62%	Faster void turnaround through clearer stage visibility and fewer handoff delays
GBP 2.4M	Projected annual saving from rent-loss prevention, compliance risk reduction, and workflow efficiency
78%	Compliance fine risk reduction through expiry visibility, escalation, and status accountability
4 roles	One platform model that still respects role-specific goals, language, and operating rhythms

Product Design Takeaway and Learnings

The strongest product decision was treating KALIK as an adaptive operating layer rather than a single dashboard. By connecting role-specific dashboards to one shared data and activity model, the design created a platform that could support executive oversight, day-to-day operations, compliance accountability, and resident-impacting repairs without fragmenting the experience again.